

Synthesis and Functionalization of ortho-Tetrafluorinated Azobenzenes for Their Incorporation into Molecular Whirligigs

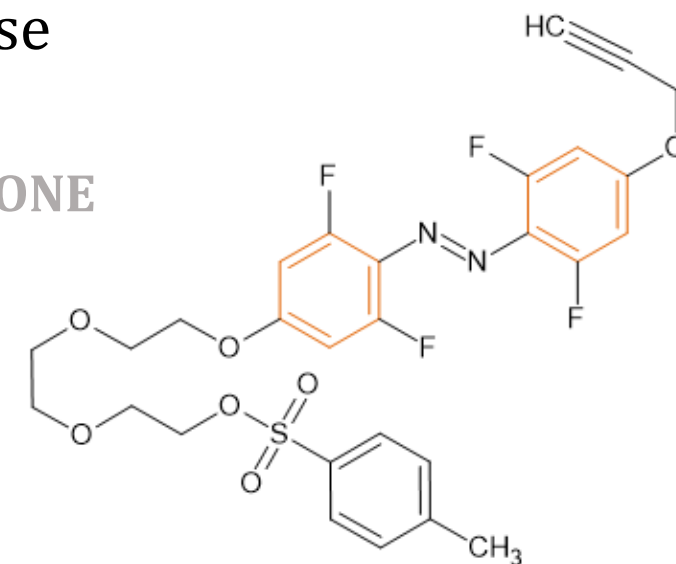
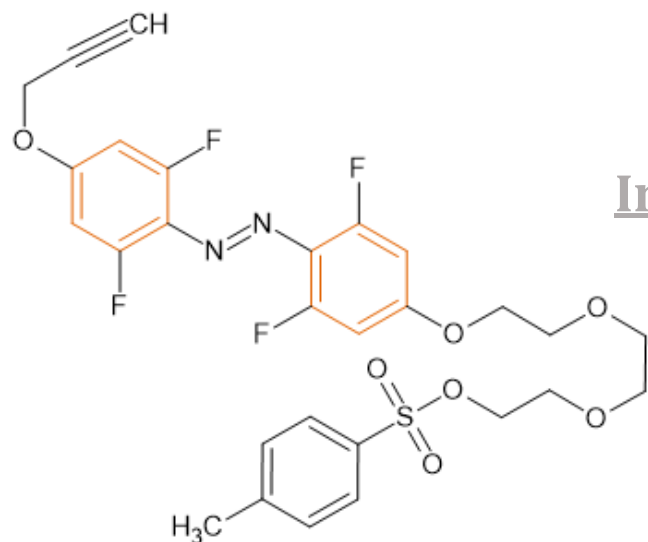
June 23, 2026

Third-Year BSc Chemistry Internship Defense

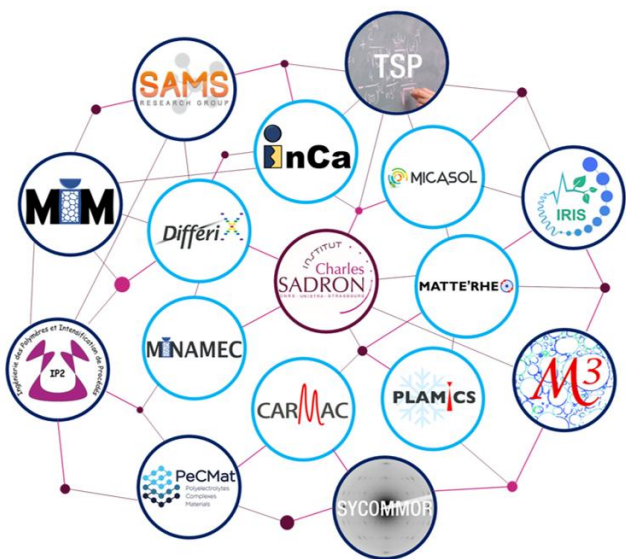
Internship supervisor: Professor Nicolas GIUSEPPONE

Supervisor: Marion BROBECKER

Presented by Anissa DEMIRSOY



Laboratory Overview



SAMS (Synthesis and Molecular & Supramolecular Self-Assembly):

- **Organic** synthesis
- **Supramolecular** chemistry
- Molecular machines → Use of a **Feringa-derived molecular motor**
- Self-assembled materials

Fundamental Chemistry Researcher:

- Literature **review** and theoretical studies
- **Design** of synthetic strategies
- **Analysis** and interpretation of results
- Scientific **communication** (publications, conferences)

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Article

Light-Driven Molecular Whirligig

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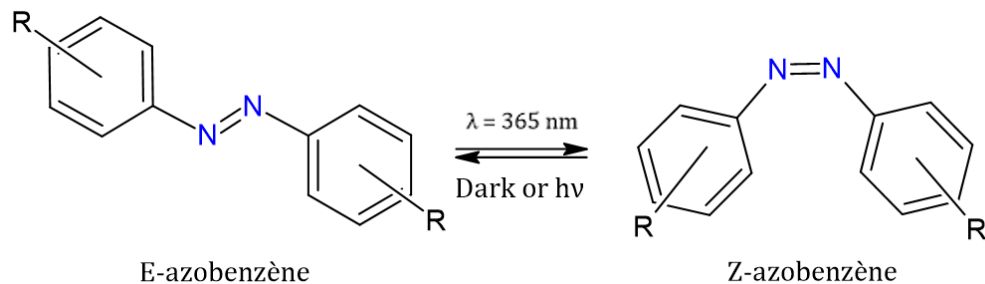


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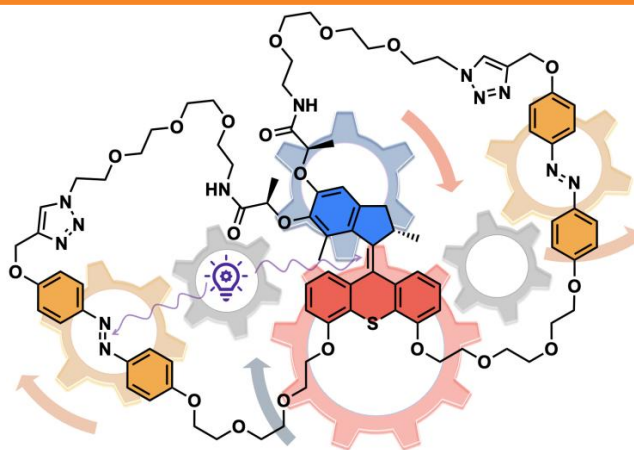
Molecular Machines

« an assembly of a distinct number of molecular components that are designed to perform machinelike movements as a result of an appropriate external stimulation » [Balzani et al.]

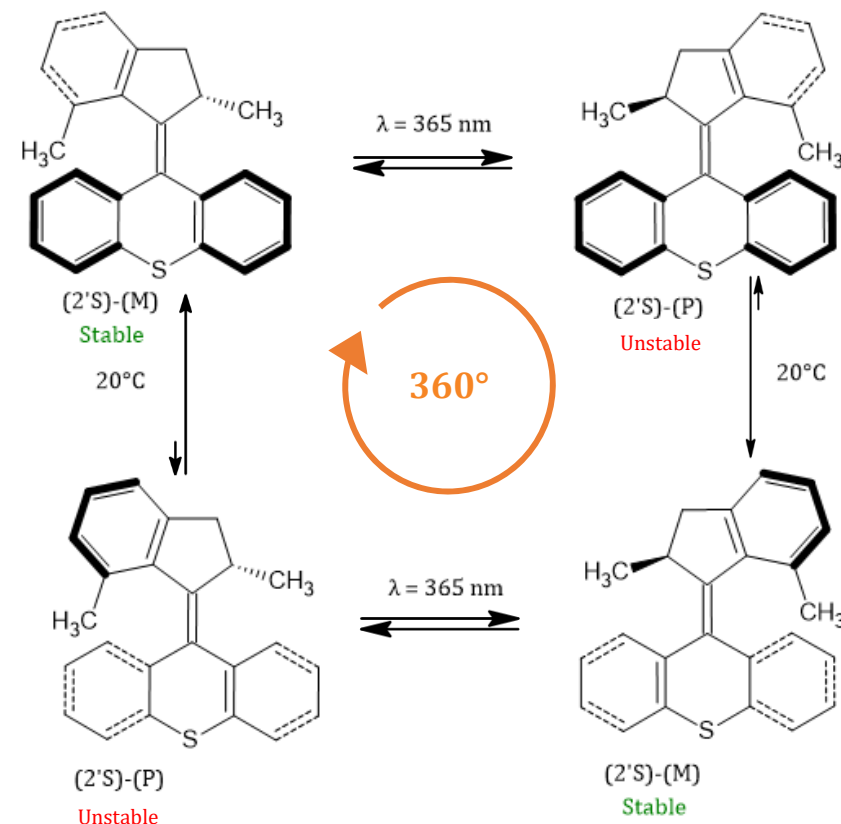
➤ Switches: **reversible** isomerization



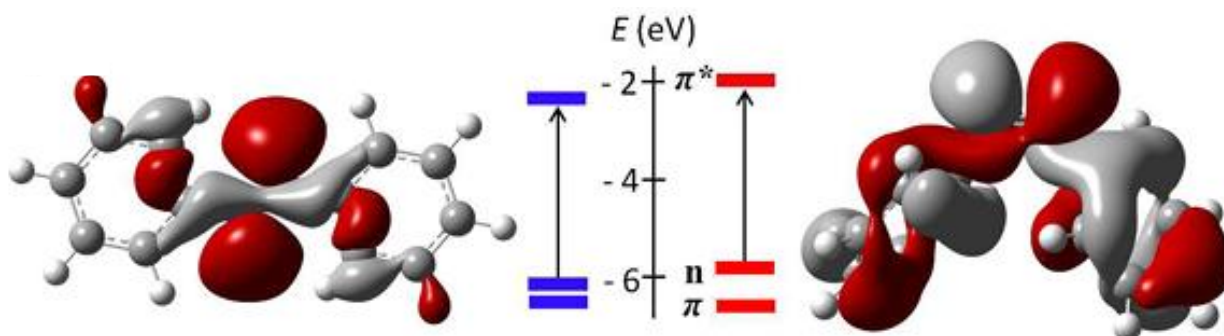
Research project: combination of azobenzene with Feringa-derived molecular motors to obtain a **molecular whirligig**



➤ Molecular motors: **unidirectional** rotation



o-Fluoroazobenzene

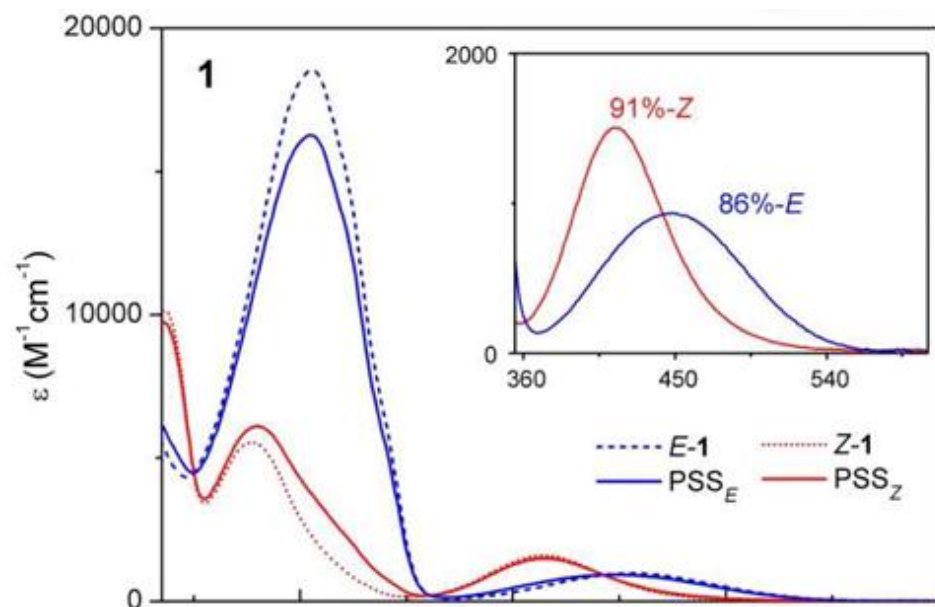


Differences in properties between azobenzene and its ortho-tetrafluorinated derivative:

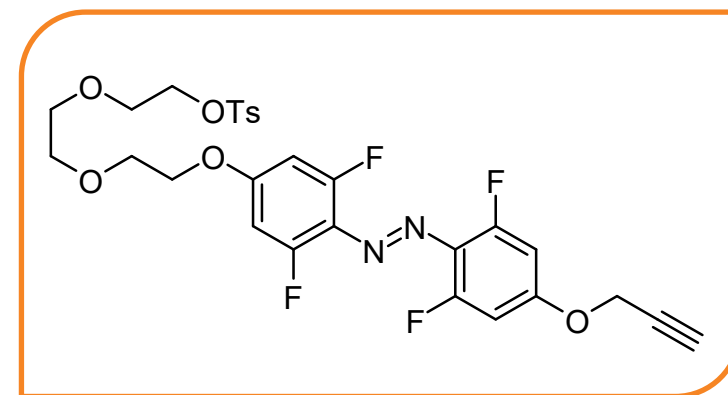
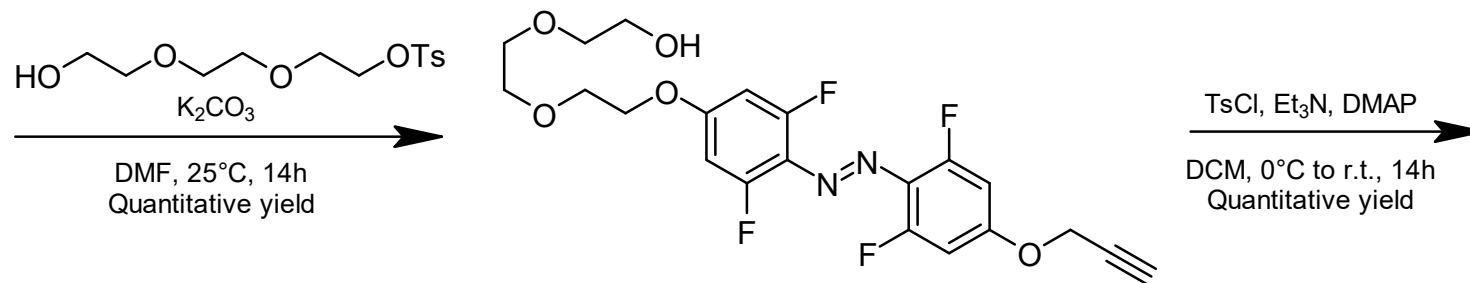
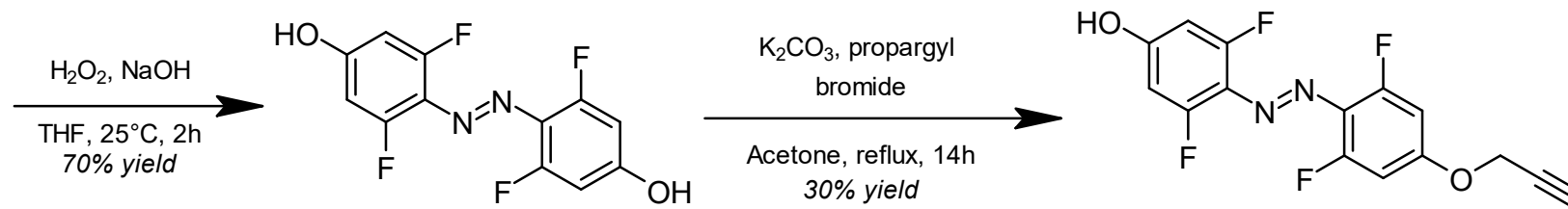
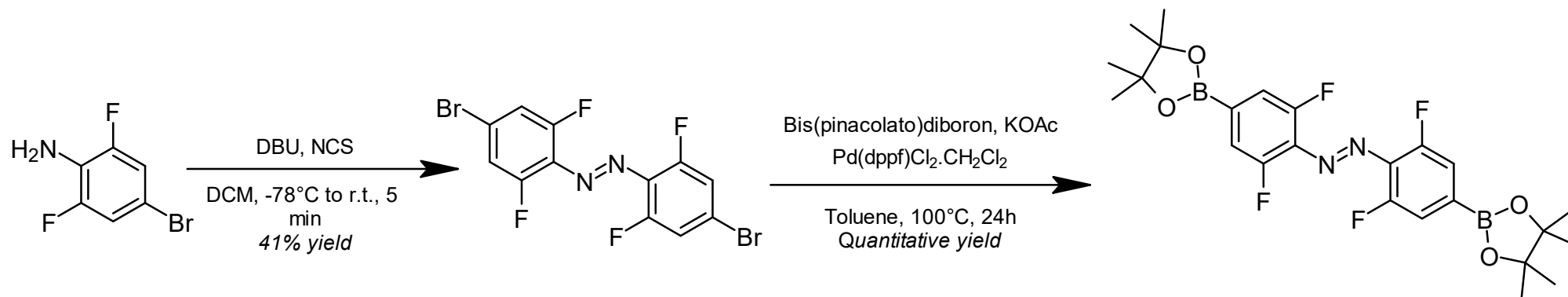
- Electron-withdrawing inductive effect of fluorine → lower E_{HOMO}
- Separation of the $n\text{-}\pi^*$ absorption bands of the E and Z isomers ($\Delta\lambda = 42 \text{ nm}$)
- $t_{1/2}(\text{Z} \rightarrow \text{E}) = 5 \text{ days}$ **vs** $t_{1/2}(\text{Z} \rightarrow \text{E}) = 700 \text{ days}$

Internship objectives:

- **Synthesis** of a disubstituted tetrafluorinated azobenzene
- Optimization of the **purification** on a gram scale
- Future **incorporation** into a molecular whirligig

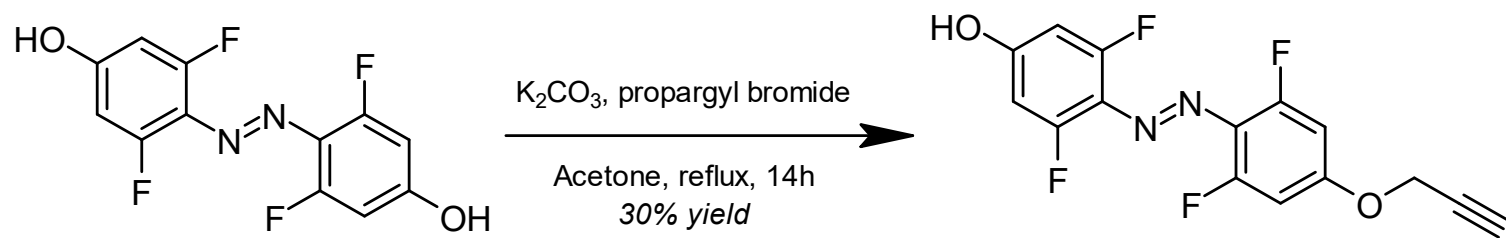


Synthetic Strategy



Monosubstitution via a Williamson Synthesis

Objective: favor the formation of pure monosubstituted azobenzene



- Synthetic strategy:
- Excess azobenzene
- Weak base (mild conditions)
- Slow dropwise addition of a dilute propargyl bromide solution

➔ **Preferential** formation of the monosubstituted product



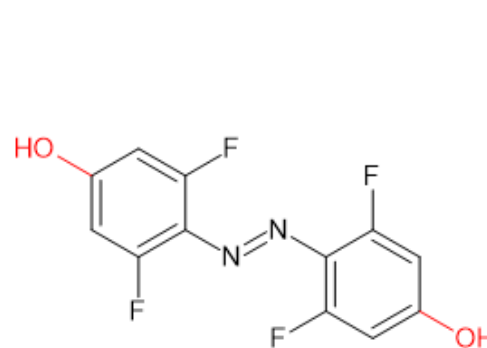
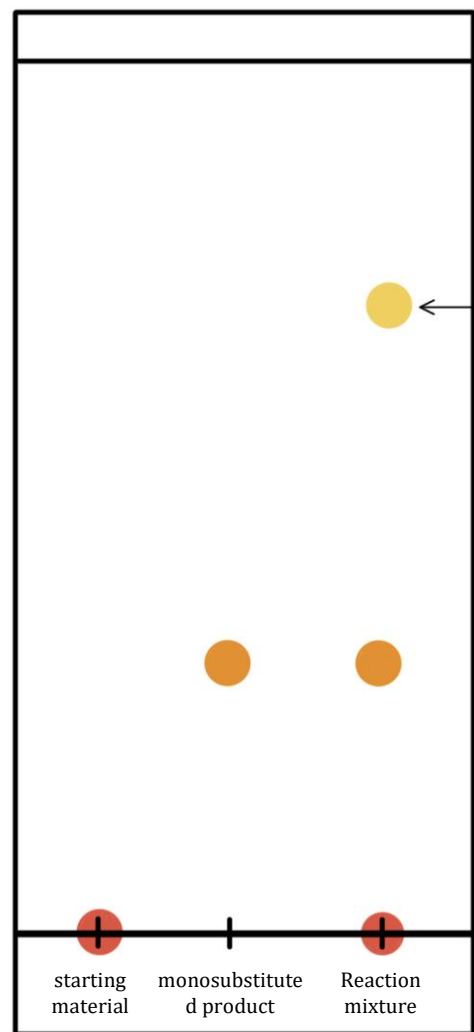
Setup under an inert atmosphere (argon)

Monosubstitution via a Williamson Synthesis

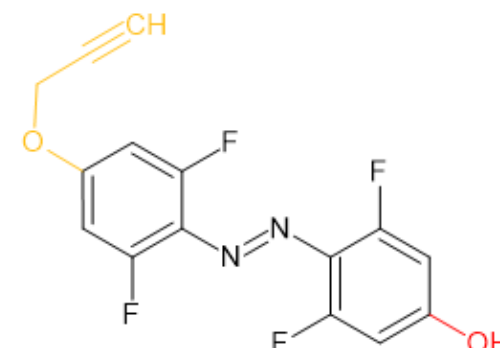
Purification Optimization:

- Preliminary study by thin-layer chromatography (TLC)
- Elution gradient:
 - Toluene (100%) → disubstituted product
 - Toluene/ethyl acetate (96:4) → monosubstituted product
 - Toluene/ ethyl acetate (70:30) → starting material

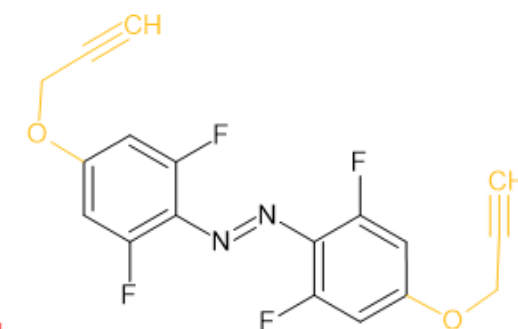
➔ Recovery of the starting material for reuse



Starting material
(*excess*)



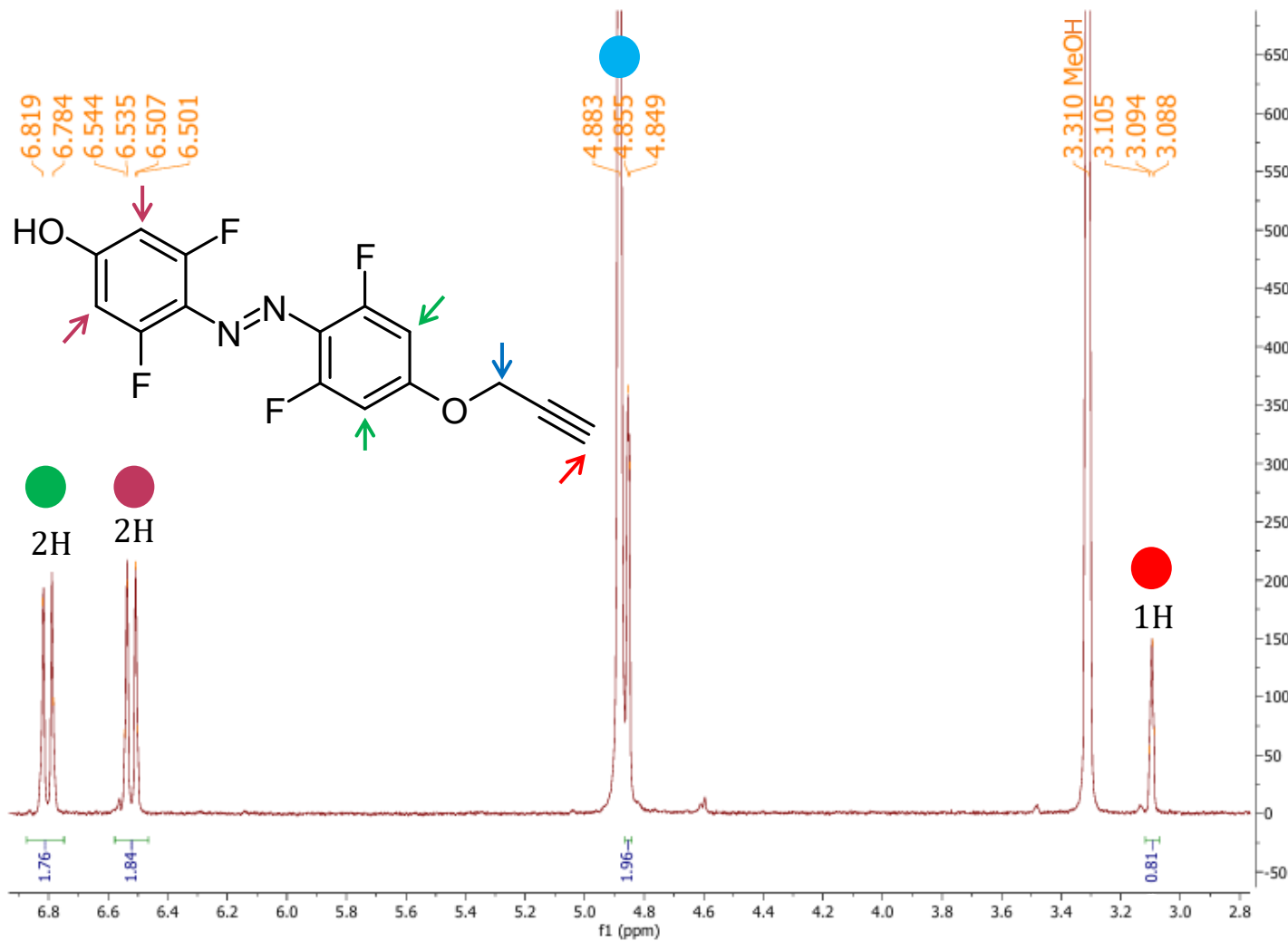
Monosubstituted
product (*major product*)



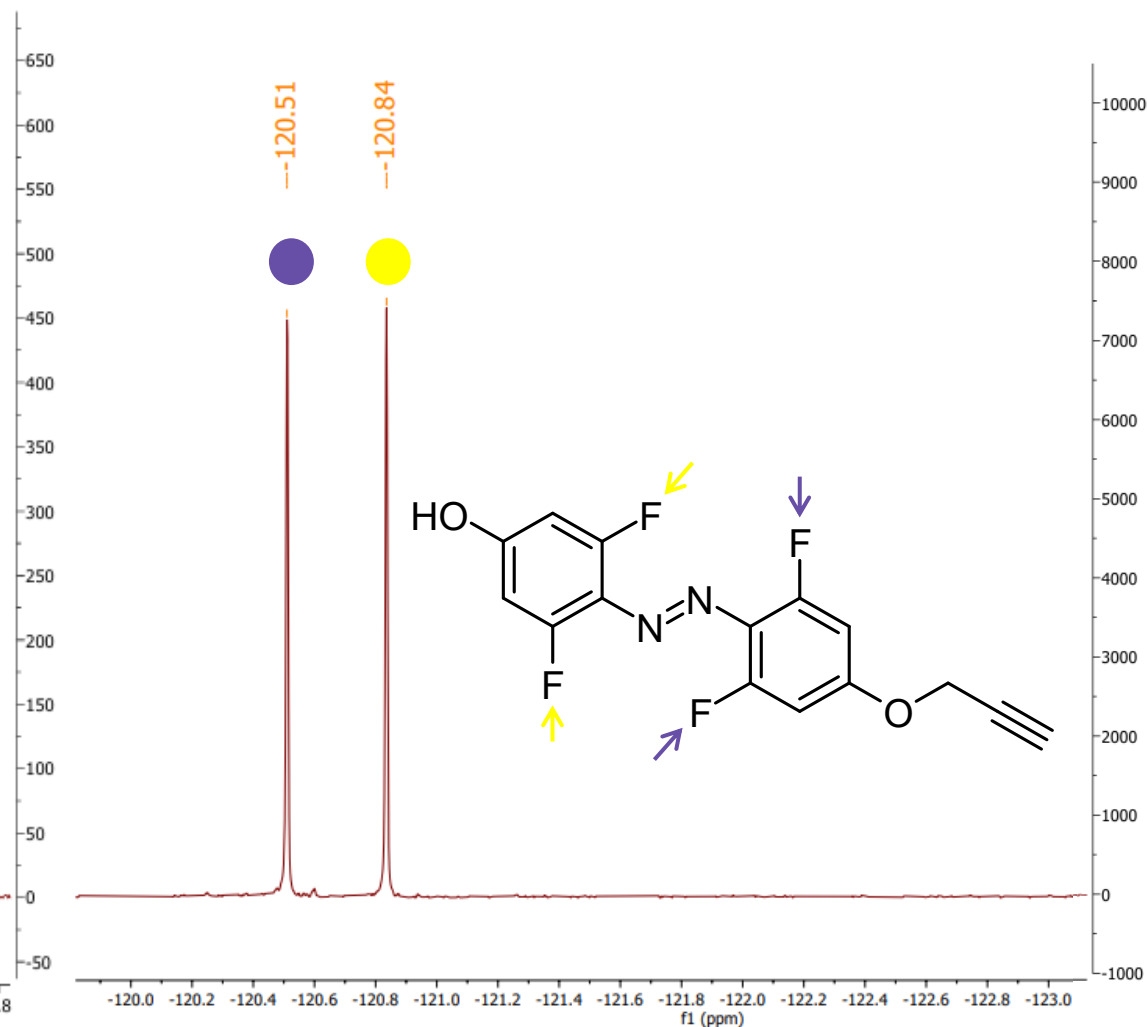
Disubstituted product
(*minor product*)

NMR Characterization

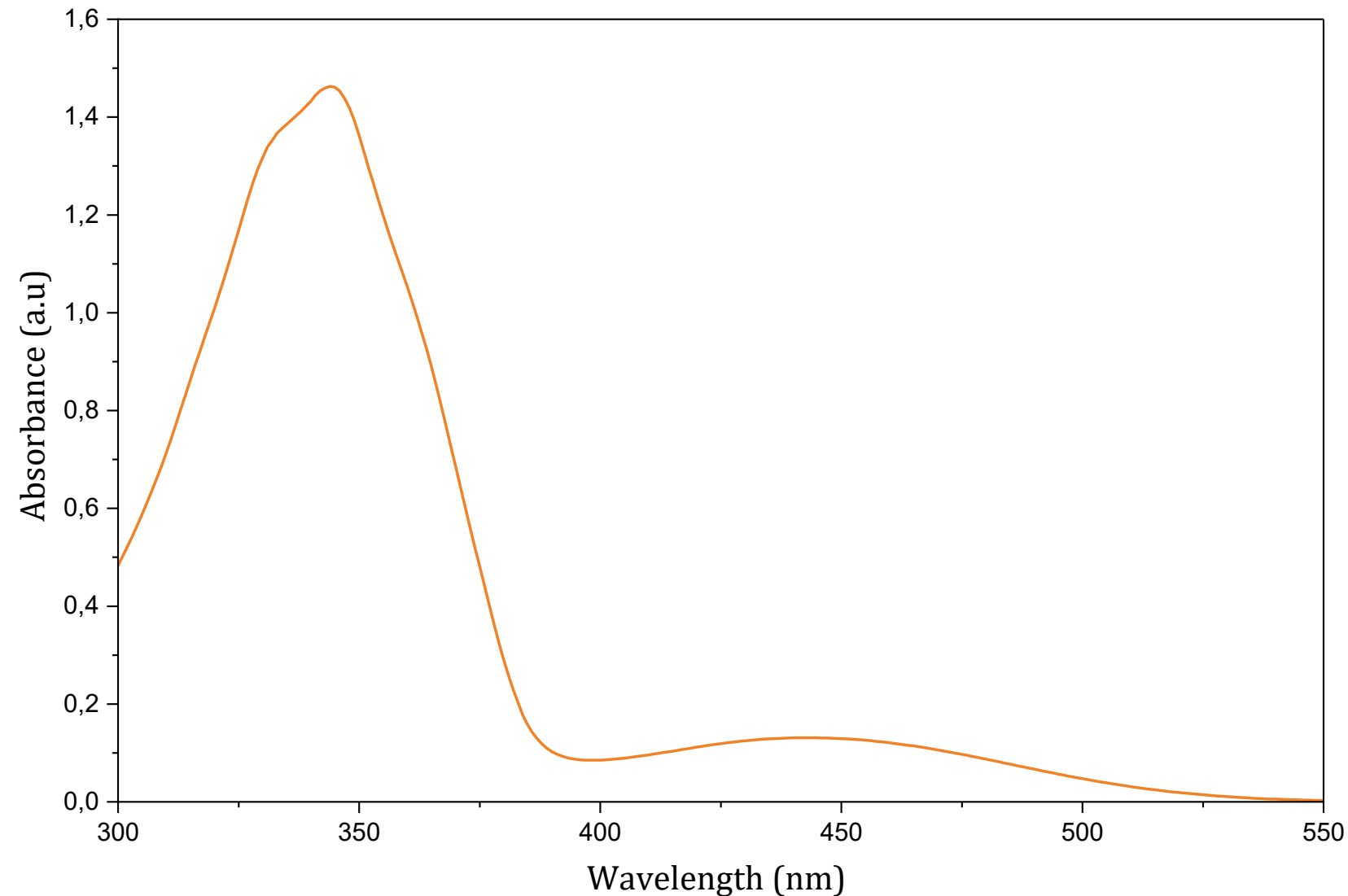
^1H NMR spectrum, solvent : CD_3OD , 400 MHz



^{19}F NMR spectrum, solvent : CD_3OD , 400 MHz



UV-Visible Spectroscopy



Absorption band at $\lambda_{\max} \simeq 345$ nm:

- $\pi \rightarrow \pi^*$
- « Allowed » transition for the **E isomer**
- **High ϵ** ($\simeq 25\,000 \text{ L}\cdot\text{mol}^{-1}\cdot\text{cm}^{-1}$)

Absorption band at $\lambda_{\max} \simeq 445$ nm:

- $n \rightarrow \pi^*$
- « Forbidden » transition for the **E isomer**
- **Low ϵ** ($\simeq 5\,000 \text{ L}\cdot\text{mol}^{-1}\cdot\text{cm}^{-1}$)

Conclusion and Perspectives

Summary:

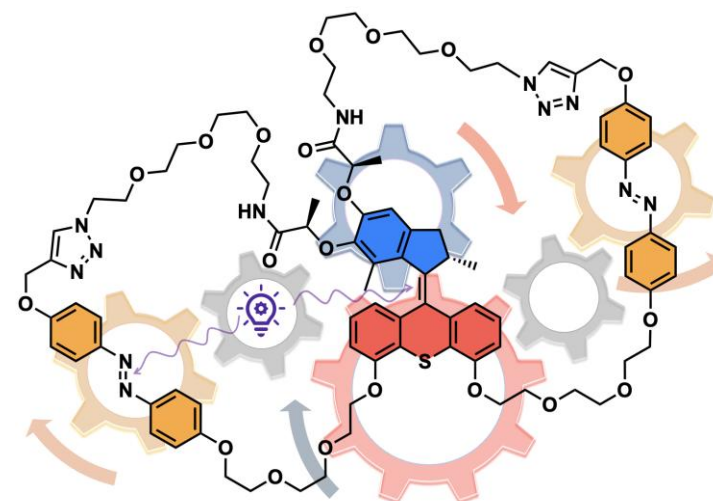
- Gram-scale synthesis of **ortho-tetrafluorinated azobenzene**
- **Optimization** of the Williamson reaction product purification by silica gel column chromatography

Perspectives:

- **Incorporation** of the azobenzene into the same system as the **Feringa-derived molecular motor**
- Future study of the competition between the molecular switch and molecular motor ($h\nu = 365 \text{ nm}$)

Personal Assessment:

- **Skills under development:** scientific English (C1), theoretical knowledge, technical skills
- **Confirmed career plan:** LPCNO internship (CNRS) → SAMS internship (CNRS) → Master's in Molecular and Supramolecular Chemistry (Unistra)



➔ **Objective: Fundamental Research**